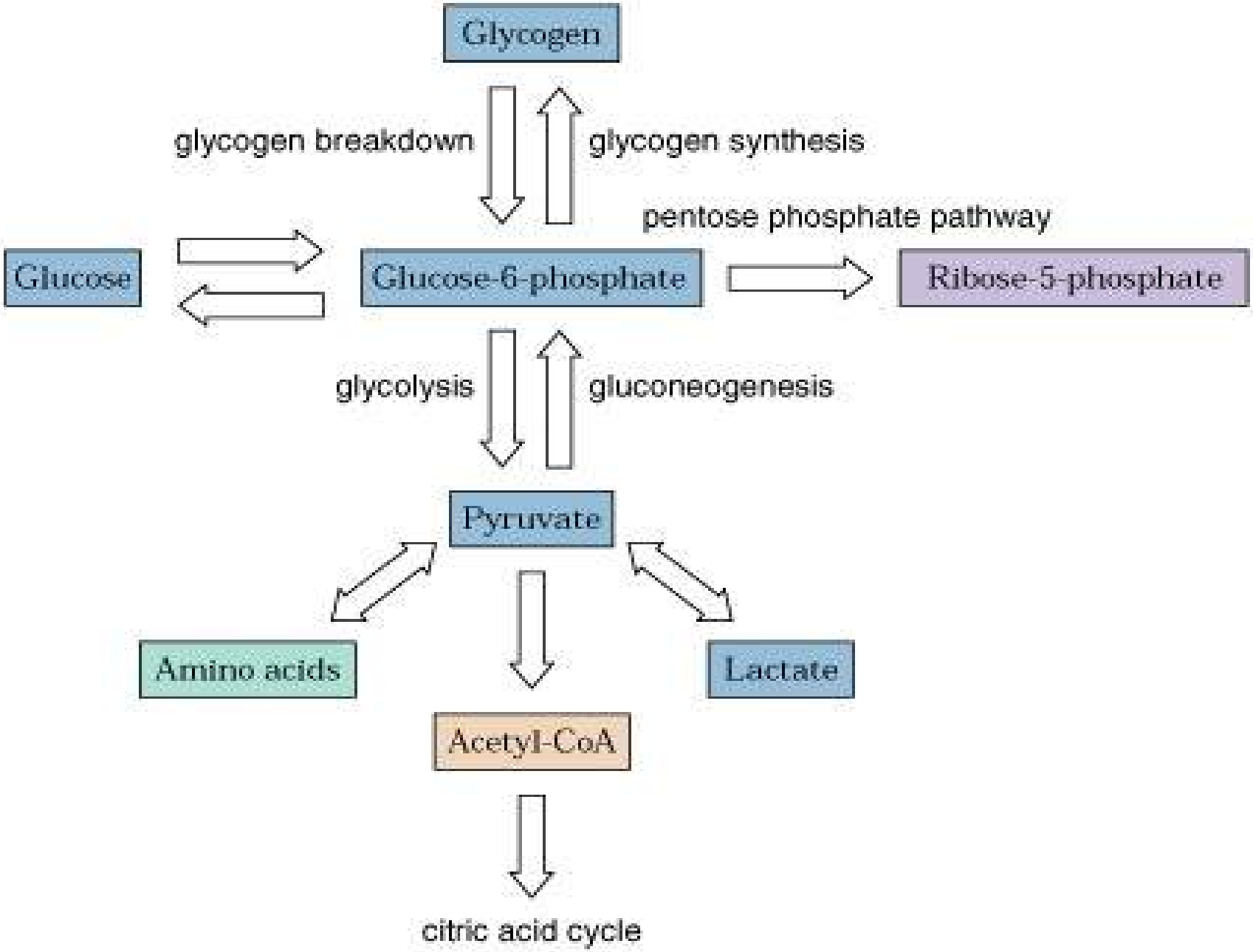
GLUCONEOGENESIS

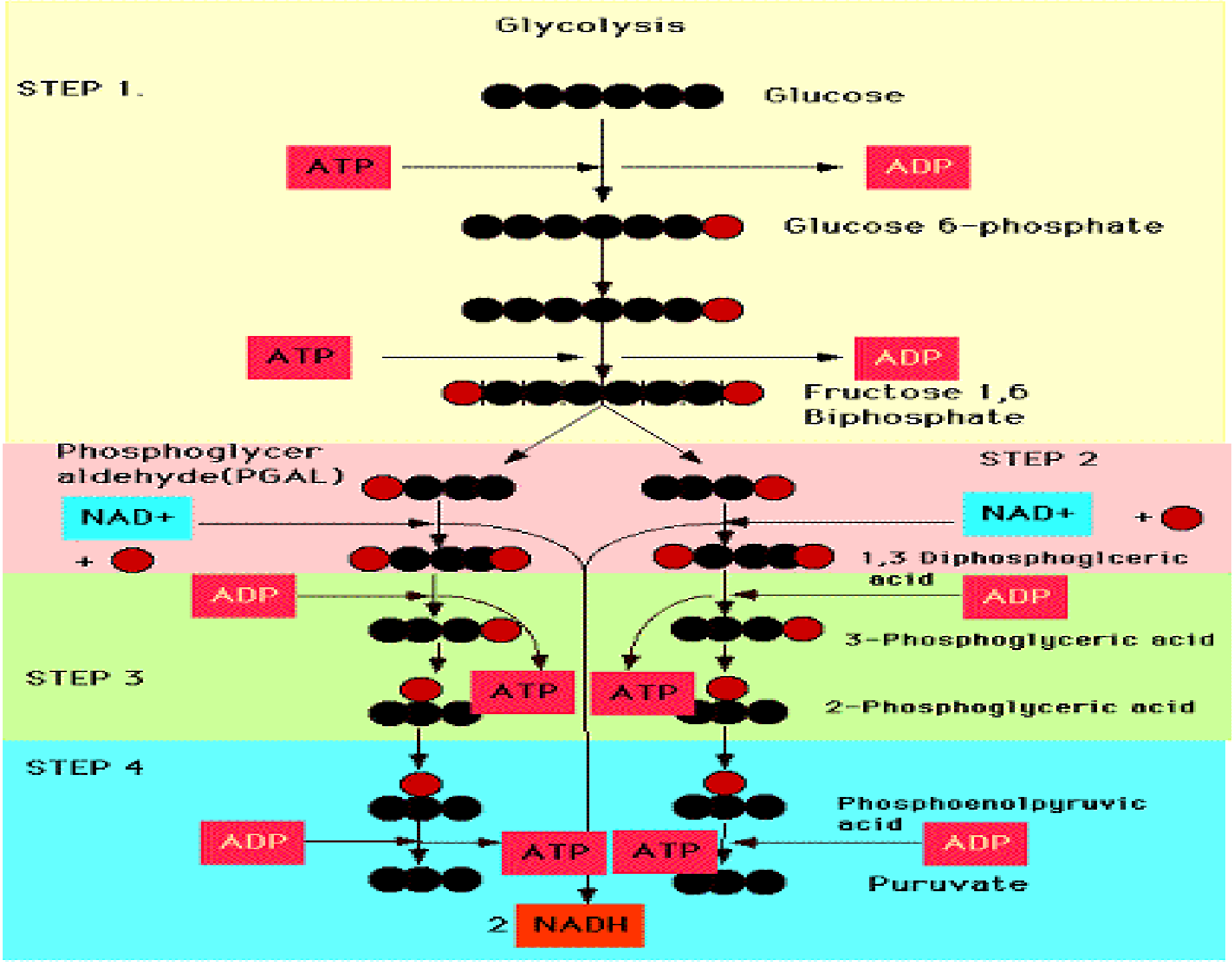


CREATING NEW GLUCOSE

Mehnaz Gitay



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Glycolysis
STEP 1.
Glucose
ATP
ADP
Glucose 6-phosphate
ATP
ADP
Fructose 1,6
Biphosphate
Phosphoglycer
aldehyde(PGAL)
STEP 2
NAD+
NAD+
+
+
1,3 Diphosphoglceric
acid
ADP
ADP
3-Phosphoglyceric acid
STEP 3
ATP
ATP
2-Phosphoglyceric acid
STEP 4
Phosphoenolpyruvic
acid
ADP
ATP
ATP
ADP
Puruvate
2
NADH

Substrates for gluconeogenesis

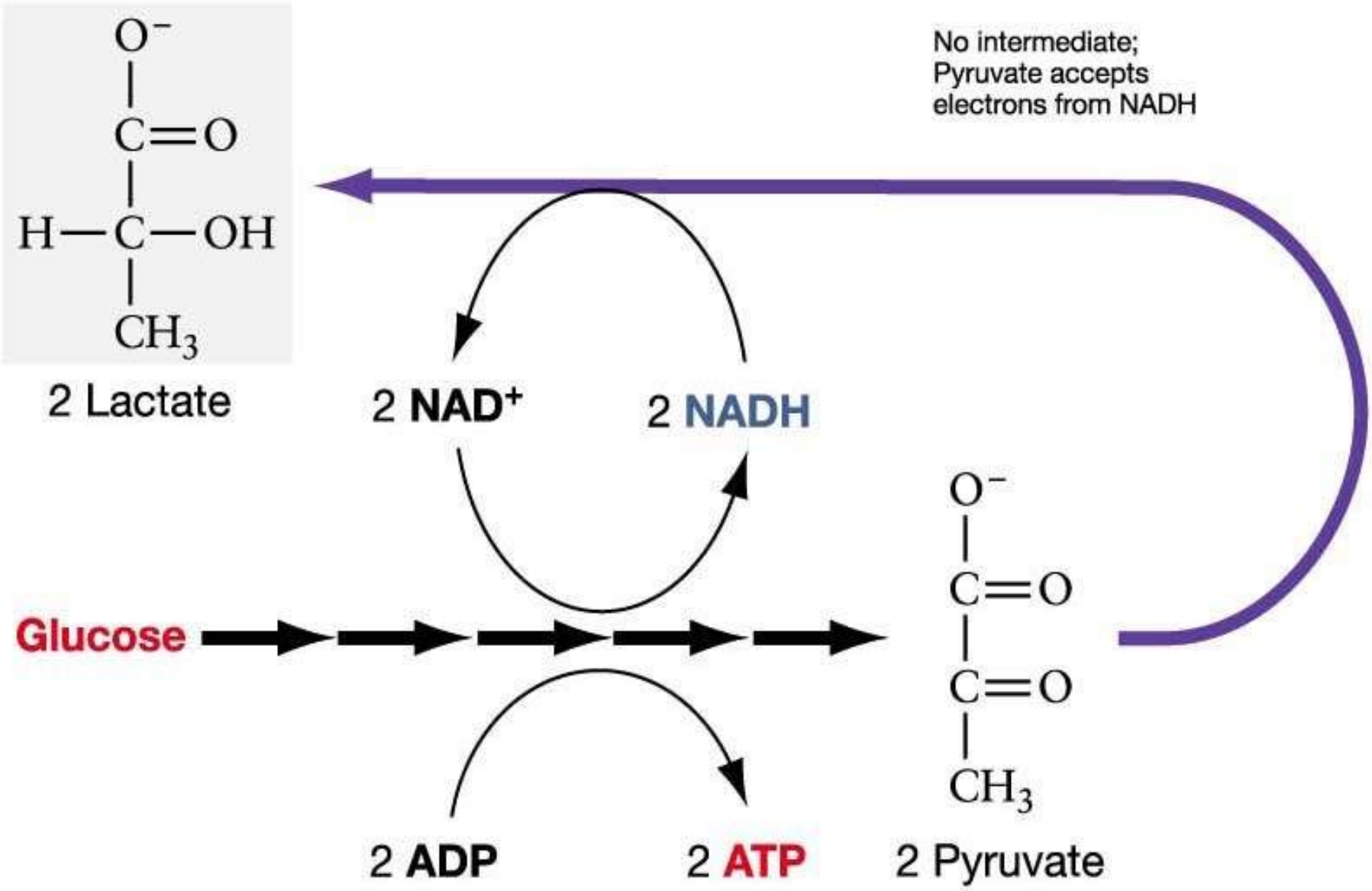
- Takes place in liver and kidney

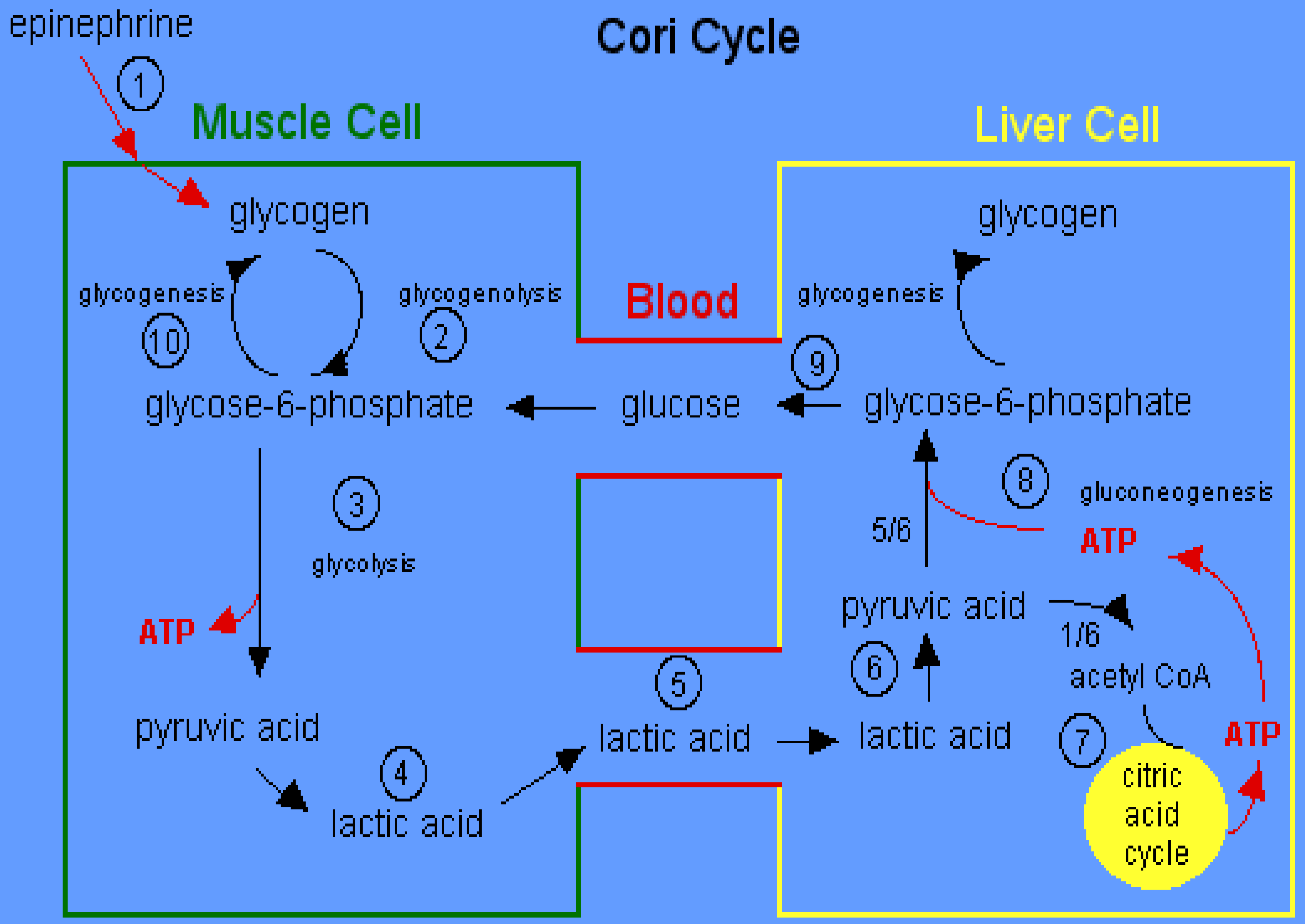
- Glycerol
- Lactate
- Amino Acids
- Propionic acid in ruminants (propionic CoA in humans,enters GNG at the point of succinyl CoA)

Glycerol

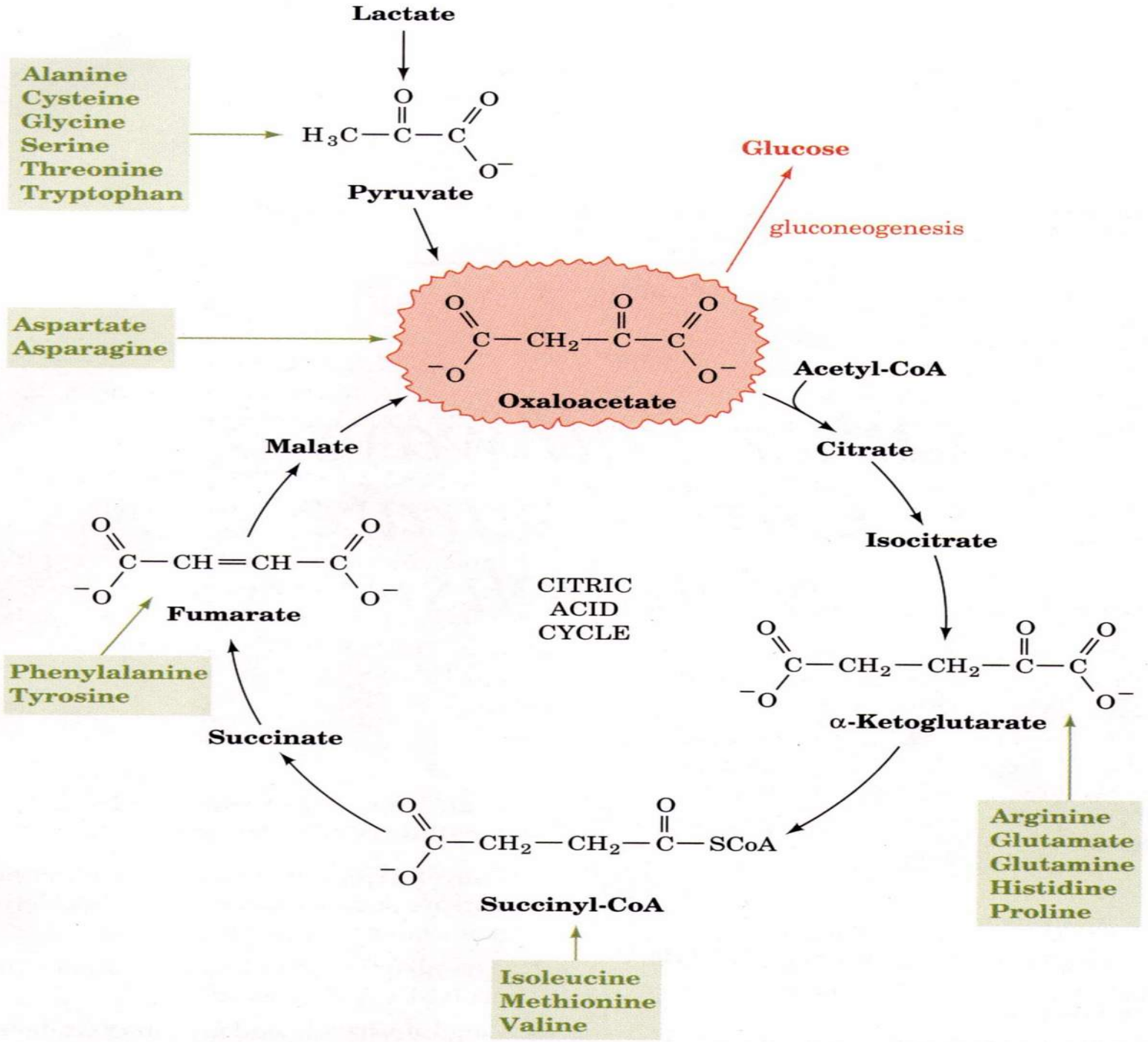
- Triacylglycerol → metabolized → Glycerol+F.A
- Glycerol → phosphorylated by enz glycerol kinase…after being transported to the liver via blood,b/c adipose tissues lack the enz.
- Glycerol phosphate is then oxidized to **Dihyroxyacetone phosphate (DHAP)** by the enz

Glycerol phosphate dehydrogenase. DHAP is an intermediate of glycolysis.


No intermediate;
Pyruvate accepts
electrons from NADH
2 Lactate
2 NAD+
2 NADH
Glucose
2 ADP
2 ATP
2 Pyruvate



C. Ophardt, c. 2003


Lactate
Alanine
Cysteine
Glycine
Serine
Threonine
Tryptophan
Pyruvate
Glucose
gluconeogenesis
Aspartate
Asparagine
Oxaloacetate
Acetyl-CoA
Malate
Citrate
Isocitrate
CITRIC
ACID
CYCLE
Fumarate
Phenylalanine
Tyrosine
α-Ketoglutarate
Succinate
Arginine
Glutamate
Glutamine
Histidine
Proline
Succinyl-CoA
Isoleucine
Methionine
Valine

Amino Acids

- Proteins breakdown → amino acids
- A-ketoglutaric acid and oxaloacetate (α- keto acids) are formed from glucogenic amino acids
- These can either enter TCA cycle or serve for gluconeogenesis
- Amino acids that form acetyl CoA cannot form glucose because conversion of acetyl CoA to pyruvate is not possible.
- Such amino acids form ketone bodies,so are ketogenic (lysine,leucine)

From Oxaloacetate to Glucose
Reverse glycolysis

- Step 1
- Carboxylation of pyruvate (mitochondrial matrix)
  - **Phosphoenol pyruvate** →irreversible → Pyruvate (3 carbon)
  - Pyruvate (3 carbon) → Carboxylation by **pyruvate carboxylase**

↓

OXALOACETATE (4 carbon)

Inner membrane impermeable

Oxaloacetate has to be transported to the cytosol for reverse glycolysis

Shuttle required to carry it out

Pyruvate carboxylase

Biotin is required as a coenzyme

- Covalently bound form of biotin to lysine residue in the enzyme is called biocytin
- High levels of acetyl CoA call for glucose formation, mostly during fasting.(Acetyl CoA build up means utilizable energy can be obtained from it whereas pyruvate should be converted back to glucose.
- Low levels of Acetyl CoA cause the TCA cycle to move on for more energy (Inactive pyruvate carboxylase)

Shuttle system

- The **malate-aspartate shuttle** (sometimes also the **malate shuttle**) is a biochemical system for translocating electrons produced during glycolysis across the semipermeable inner membrane of the mitochondrion for oxidative phosphorylation in eukaryotes.

How does it function?

- Enzymes
  - mitchondrial malate dehydrogenase and cytosolic malate dehydrogenase
  - Oxaloacetate is reduced to form malate and NAD^+ by the enz m.malate dehydrogenase.
  - Inner membrane permeable to malate, which is transported out in the cytosol
  - Cytosolic malate dehydrogenase again dehydrogenates malate via NAD^+ to form oxaloacetate
  - The antiporter aspartate and glutamate is involved the reaction being through aminotransferase, thus called MALATE ASPARTATE SHUTTLE

Oxaloacetate in the cytosol

- Oxaloacetate in the cytosol is converted to Phosphoenol pyruvate (PEP), enz Phosphoenol pyruvate carboxykinase (PEPCK). The PEPCK enz is also found in mitochondria in humans. The PEP formed in mitochondria is transported by a special transporter to the cytosol.
- Oxaloaceate → decarboxylation and phosphorylation

  ↓ GTP → GDP + Pi

  Phosphoenolpyruvate

Formation of Fructose -6-PO4

- Enzyme

  Fructose -1,6 bisphosphatase

The enzyme is inhibited by high levels of AMP (energy poor state) which guides more formation of ATP.

Formation of Glucose

- Enzyme

  Glucose -6- phosphatase(only in liver and kidney, works together with Glucose -6- phospate translocase)

Translocase transports the glucose 6 phosphate across the endoplasmic reticular membrane.

Forms free glucose

Lack of the above enzymes render the muscle glycogen unable to effect the blood pool of glucose.

- Special transporters then transfer free glucose and phosphate from the ER membrane to the cytosol.

Summary

- Mitochondria---- pyruvate → pyruvate carboxylase
- → oxaloacetate → shuttle → cytosol → phosphoenolpyruvate carboxykinase (PEPCK) → PEP → Reverse glycolysis → Fructose 1,6 bisphosphatase → fructose 6 phosphate → reverse glycolysis → glucose 6 phosphatase → GLUCOSE
- Hormones
- Glucagon → GNG from lactic acid and amino acids
- Glucocorticoids → Protein catabolism and Amino acid uptake

Ir-reversible reactions of glycolysis bypassed

- Pyruvate kinase → pyruvate carboxylase & PEPCK
- Phosphofructokinase 1 → Fructose 1,6 bisphosphatase
- Hexokinase → Glucose phosphatase

Reading



- Chatterjea
- Satyanarayan